

tf-idf, vector space model, and BM25

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slides borrowed from
[Stanford's CS 276 / LING 286](#)

This lecture; IIR Sections 6.2-6.4.3

- Ranked retrieval
- Scoring documents
- Term frequency
- Collection statistics
- Weighting schemes
- Vector space scoring

Ranked retrieval

- Thus far, our queries have all been Boolean.
 - Documents either match or don't.
- Good for expert users with precise understanding of their needs and the collection.
 - Also good for applications: Applications can easily consume 1000s of results.
- Not good for the majority of users.
 - Most users incapable of writing Boolean queries (or they are, but they think it's too much work).
 - Most users don't want to wade through 1000s of results.
 - This is particularly true of web search.

Problem with Boolean search: feast or famine

- Boolean queries often result in either too few (=0) or too many (1000s) results.
- Query 1: “*standard user dlink 650*” → 200,000 hits
- Query 2: “*standard user dlink 650 no card found*”: 0 hits
- It takes a lot of skill to come up with a query that produces a manageable number of hits.
 - AND gives too few; OR gives too many

Ranked retrieval models

- Rather than a set of documents satisfying a query expression, in **ranked retrieval**, the system returns an ordering over the (top) documents in the collection for a query
- **Free text queries**: Rather than a query language of operators and expressions, the user's query is just one or more words in a human language
- In principle, there are two separate choices here, but in practice, ranked retrieval has normally been associated with free text queries and vice versa

Feast or famine: not a problem in ranked retrieval

- When a system produces a ranked result set, large result sets are not an issue
 - Indeed, the size of the result set is not an issue
 - We just show the top k (≈ 10) results
 - We don't overwhelm the user
- Premise: the ranking algorithm works

Scoring as the basis of ranked retrieval

- We wish to return in order the documents most likely to be useful to the searcher
- How can we rank-order the documents in the collection with respect to a query?
- Assign a score – say in $[0, 1]$ – to each document
- This score measures how well document and query “match”.

Query-document matching scores

- We need a way of assigning a score to a query/document pair
- Let's start with a one-term query
- If the query term does not occur in the document: score should be 0
- The more frequent the query term in the document, the higher the score (should be)
- We will look at several alternatives for this.

Binary term-document incidence matrix

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	1	1	0	0	0	1
Brutus	1	1	0	1	0	0
Caesar	1	1	0	1	1	1
Calpurnia	0	1	0	0	0	0
Cleopatra	1	0	0	0	0	0
mercy	1	0	1	1	1	1
worser	1	0	1	1	1	0

Each document is represented by a binary vector $\in \{0,1\}^{|M|}$

Term-document count matrices

- Consider the number of occurrences of a term in a document:
 - Each document is a **count vector** in $\mathbb{N}^v$: a column below

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	157	73	0	0	0	0
Brutus	4	157	0	1	0	0
Caesar	232	227	0	2	1	1
Calpurnia	0	10	0	0	0	0
Cleopatra	57	0	0	0	0	0
mercy	2	0	3	5	5	1
worser	2	0	1	1	1	0

Bag of words model

- Vector representation doesn't consider the ordering of words in a document
- *John is quicker than Mary* and *Mary is quicker than John* have the same vectors
- This is called the bag of words model.
- In a sense, this is a step back: The positional index was able to distinguish these two documents.

Term frequency tf

- The term frequency $tf_{t,d}$ of term t in document d is defined as the number of times that t occurs in d .
 - Note: Frequency means count in IR
- We want to use tf when computing query-document match scores. But how?
- Raw term frequency is not what we want:
 - A document with 10 occurrences of the term is more relevant than a document with 1 occurrence of the term.
 - But not 10 times more relevant.
- Relevance does not increase proportionally with term frequency.

Log-frequency weighting

- The log frequency weight of term t in d is

$$w_{t,d} = \begin{cases} 1 + \log_{10} \text{tf}_{t,d}, & \text{if } \text{tf}_{t,d} > 0 \\ 0, & \text{otherwise} \end{cases}$$

- $0 \rightarrow 0, 1 \rightarrow 1, 2 \rightarrow 1.3, 10 \rightarrow 2, 1000 \rightarrow 4$, etc.
- Score for a document-query pair: sum over terms t in both q and d :
- score
$$= \sum_{t \in q \cap d} (1 + \log \text{tf}_{t,d})$$
- The score is 0 if none of the query terms is present in the document.

Rare terms are more informative

- Rare terms are more informative than frequent terms
 - Recall stop words
- Consider a term in the query that is rare in the collection (e.g., *arachnocentric*)
- A document containing this term is very likely to be relevant to the query *arachnocentric*
- → We want a high weight for rare terms like *arachnocentric*.

Collection vs. Document frequency

- Collection frequency of t is the number of occurrences of t in the collection
- Document frequency of t is the number of documents in which t occurs
- Example:

Word	Collection frequency	Document frequency
<i>insurance</i>	10440	3997
<i>try</i>	10422	8760

- Which word is for better search (gets higher weight)

idf weight

- df_t is the document frequency of t : the number of documents that contain t
 - df_t is an inverse measure of the informativeness of t
 - $df_t \leq N$
- We define the idf (inverse document frequency) of t by
$$idf_t = \log_{10} (N/df_t)$$
 - We use $\log (N/df_t)$ instead of N/df_t to “dampen” the effect of idf.

idf example, suppose $N = 1$ million

term	df_t	idf_t
calpurnia	1	6
animal	100	4
sunday	1,000	3
fly	10,000	2
under	100,000	1
the	1,000,000	0

$$idf_t = \log_{10} (N/df_t)$$

There is one idf value for each term t in a collection.

Effect of idf on ranking

- Does idf have an effect on ranking for one-term queries, like
 - iPhone
- idf has no effect on ranking one term queries
 - idf affects the ranking of documents for queries with at least two terms
- For the query capricious person, idf weighting makes occurrences of capricious count for much more in the final document ranking than occurrences of person.

tf-idf weighting

- The tf-idf weight of a term is the product of its tf weight and its idf weight.

$$w_{t,d} = \log(1 + \text{tf}_{t,d}) \times \log_{10}(N / \text{df}_t)$$

- **Best known weighting scheme in information retrieval**
 - Note: the “-” in tf-idf is a hyphen, not a minus sign!
 - **Alternative names: tf.idf, tf x idf**
- Increases with the number of occurrences within a document
- **Increases with the rarity of the term in the collection**

Score for a document given a query

$$\text{Score}(q, d) = \sum_{t \in q \cap d} \text{tf.idf}_{t,d}$$

- There are many variants
 - How “tf” is computed (with/without logs)
 - Whether the terms in the query are also weighted
 - ...

Binary $\rightarrow$ count $\rightarrow$ weight matrix

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	5.25	3.18	0	0	0	0.35
Brutus	1.21	6.1	0	1	0	0
Caesar	8.59	2.54	0	1.51	0.25	0
Calpurnia	0	1.54	0	0	0	0
Cleopatra	2.85	0	0	0	0	0
mercy	1.51	0	1.9	0.12	5.25	0.88
worser	1.37	0	0.11	4.15	0.25	1.95

Each document is now represented by a real-valued vector of tf-idf weights $\in \mathbb{R}^{|V|}$

Documents as vectors

- So we have a $|V|$ -dimensional vector space
- Terms are axes of the space
- Documents are points or vectors in this space
- Very high-dimensional: tens of millions of dimensions when you apply this to a web search engine
- These are very sparse vectors - most entries are zero.

Queries as vectors

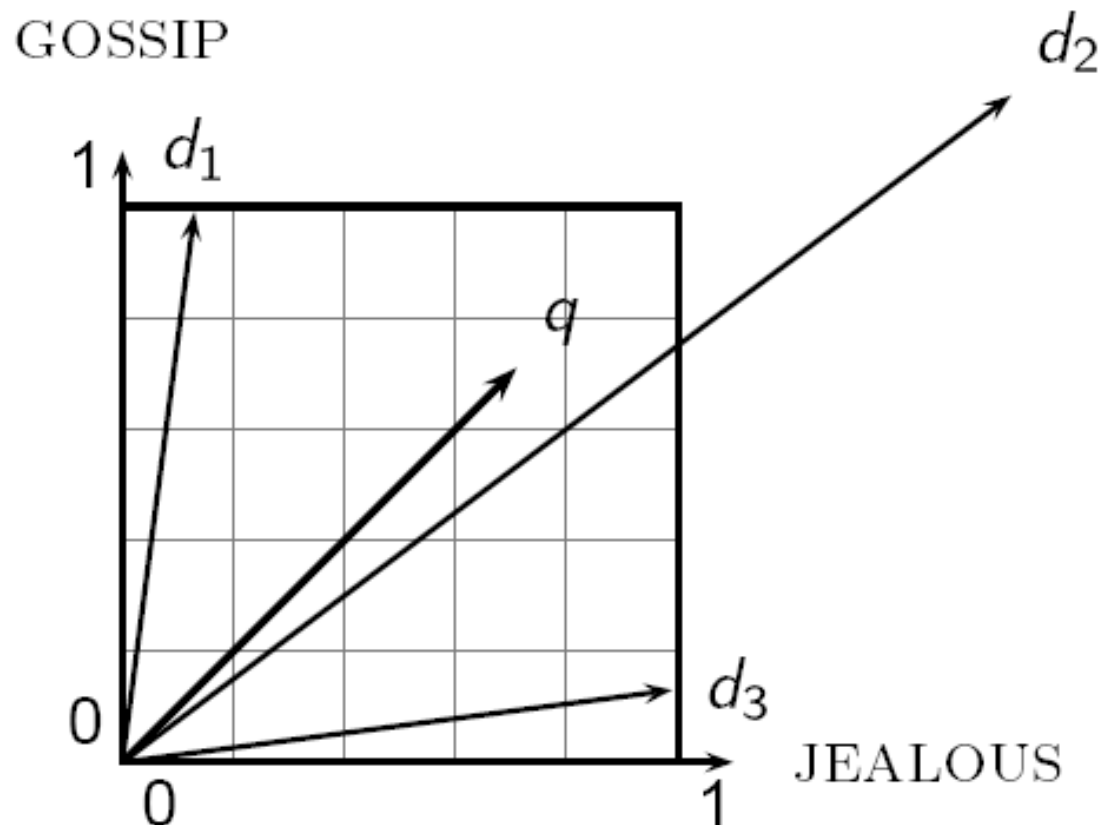
- Key idea 1: Do the same for queries: represent them as vectors in the space
- Key idea 2: Rank documents according to their proximity to the query in this space
- proximity = similarity of vectors
- proximity $\approx$ inverse of distance

Formalizing vector space proximity

- First cut: distance between two points
 - (= distance between the end points of the two vectors)
- **Euclidean distance?**
- Euclidean distance is a bad idea . . .
- . . . because Euclidean distance is **large** for vectors of **different lengths**.

Why distance is a bad idea

The Euclidean distance between $\vec{q}$ and $\vec{d}_2$ is large even though the distribution of terms in the query $\vec{q}$ and the distribution of terms in the document $\vec{d}_2$ are very similar.



Use angle instead of distance

- Thought experiment: take a document d and append it to itself. Call this document d' .
- “Semantically” d and d' have the same content
- The Euclidean distance between the two documents can be quite large
- The angle between the two documents is 0, corresponding to maximal similarity.
- Key idea: Rank documents according to angle with query.

From angles to cosines

- The following two notions are equivalent.
 - Rank documents in decreasing order of the angle between query and document
 - Rank documents in increasing order of $\cos(\text{query}, \text{document})$
- Cosine is a monotonically decreasing function for the interval $[0^\circ, 180^\circ]$

Length normalization

- A vector can be (length-) normalized by dividing each of its components by its length – for this we use the L_2 norm:

$$\|\vec{x}\|_2 = \sqrt{\sum_i x_i^2}$$

- Dividing a vector by its L_2 norm makes it a unit (length) vector (on surface of unit hypersphere)
- Effect on the two documents d and d' (d appended to itself) from earlier slide: they have identical vectors after length-normalization.
 - Long and short documents now have comparable weights

cosine(query,document)

$$\cos(\vec{q}, \vec{d}) = \frac{\vec{q} \bullet \vec{d}}{|\vec{q}| |\vec{d}|} = \frac{\vec{q}}{|\vec{q}|} \bullet \frac{\vec{d}}{|\vec{d}|} = \frac{\sum_{i=1}^{|V|} q_i d_i}{\sqrt{\sum_{i=1}^{|V|} q_i^2} \sqrt{\sum_{i=1}^{|V|} d_i^2}}$$

Dot product
Unit vectors

q_i is the weight of term i in the query

d_i is the weight of term i in the document

$\cos(\vec{q}, \vec{d})$ is the cosine similarity of $\vec{q}$ and $\vec{d}$... or, equivalently, the cosine of the angle between $\vec{q}$ and $\vec{d}$.

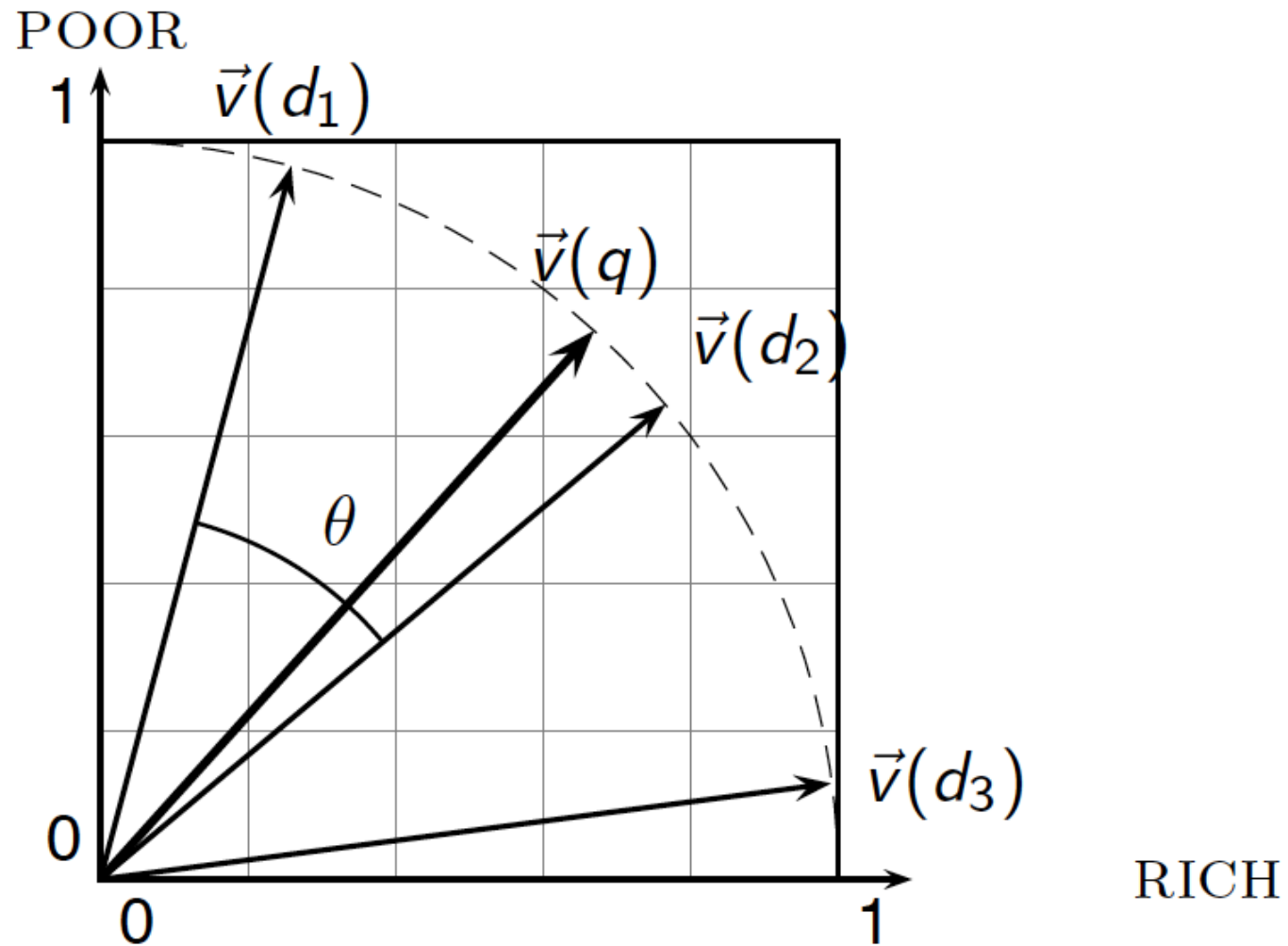
Cosine for length-normalized vectors

- For length-normalized vectors, cosine similarity is simply the dot product (or scalar product):

$$\cos(\vec{q}, \vec{d}) = \vec{q} \bullet \vec{d} = \sum_{i=1}^{|V|} q_i d_i$$

for q, d length-normalized.

Cosine similarity illustrated



Cosine similarity amongst 3 documents

How similar are
the novels

SaS: *Sense and
Sensibility*

PaP: *Pride and
Prejudice*, and

WH: *Wuthering
Heights*?

term	SaS	PaP	WH
affection	115	58	20
jealous	10	7	11
gossip	2	0	6
wuthering	0	0	38

Term frequencies (counts)

Note: To simplify this example, we don't do idf weighting.

3 documents example contd.

Log frequency weighting

term	SaS	PaP	WH
affection	3.06	2.76	2.30
jealous	2.00	1.85	2.04
gossip	1.30	0	1.78
wuthering	0	0	2.58

$$\text{dot}(\text{SaS}, \text{PaP}) \approx 12.1$$

$$\text{dot}(\text{SaS}, \text{WH}) \approx 13.4$$

$$\text{dot}(\text{PaP}, \text{WH}) \approx 10.1$$

After length normalization

term	SaS	PaP	WH
affection	0.789	0.832	0.524
jealous	0.515	0.555	0.465
gossip	0.335	0	0.405
wuthering	0	0	0.588

$$\cos(\text{SaS}, \text{PaP}) \approx 0.94$$

$$\cos(\text{SaS}, \text{WH}) \approx 0.79$$

$$\cos(\text{PaP}, \text{WH}) \approx 0.69$$

Computing cosine scores

COSINESCORE(q)

```
1  float Scores[ $N$ ] = 0
2  float Length[ $N$ ]
3  for each query term  $t$ 
4  do calculate  $w_{t,q}$  and fetch postings list for  $t$ 
5      for each pair( $d, tf_{t,d}$ ) in postings list
6      do Scores[ $d$ ] + =  $w_{t,d} \times w_{t,q}$ 
7  Read the array Length
8  for each  $d$ 
9  do Scores[ $d$ ] = Scores[ $d$ ] / Length[ $d$ ]
10 return Top  $K$  components of Scores[]
```

tf-idf example: Inc.ltc

Document: *car insurance auto insurance*

Query: *best car insurance*

Term	Query						Document				Pro d
	tf- raw	tf-wt	df	idf	wt	n'liz e	tf-raw	tf-wt	wt	n'liz e	
auto	0	0	5000	2.3	0	0	1	1	1	0.52	0
best	1	1	50000	1.3	1.3	0.34	0	0	0	0	0
car	1	1	10000	2.0	2.0	0.52	1	1	1	0.52	0.27
insurance	1	1	1000	3.0	3.0	0.78	2	1.3	1.3	0.68	0.53

$$\text{Doc length} = \sqrt{1^2 + 0^2 + 1^2 + 1.3^2} \approx 1.92$$

$$\text{Score} = 0 + 0 + 0.27 + 0.53 = 0.8$$

Summary – vector space ranking

- Represent the query as a weighted tf-idf vector
- Represent each document as a weighted tf-idf vector
- Compute the cosine similarity score for the query vector and each document vector
- Rank documents with respect to the query by score
- Return the top K (e.g., $K = 10$) to the user

BM25

Okapi BM25

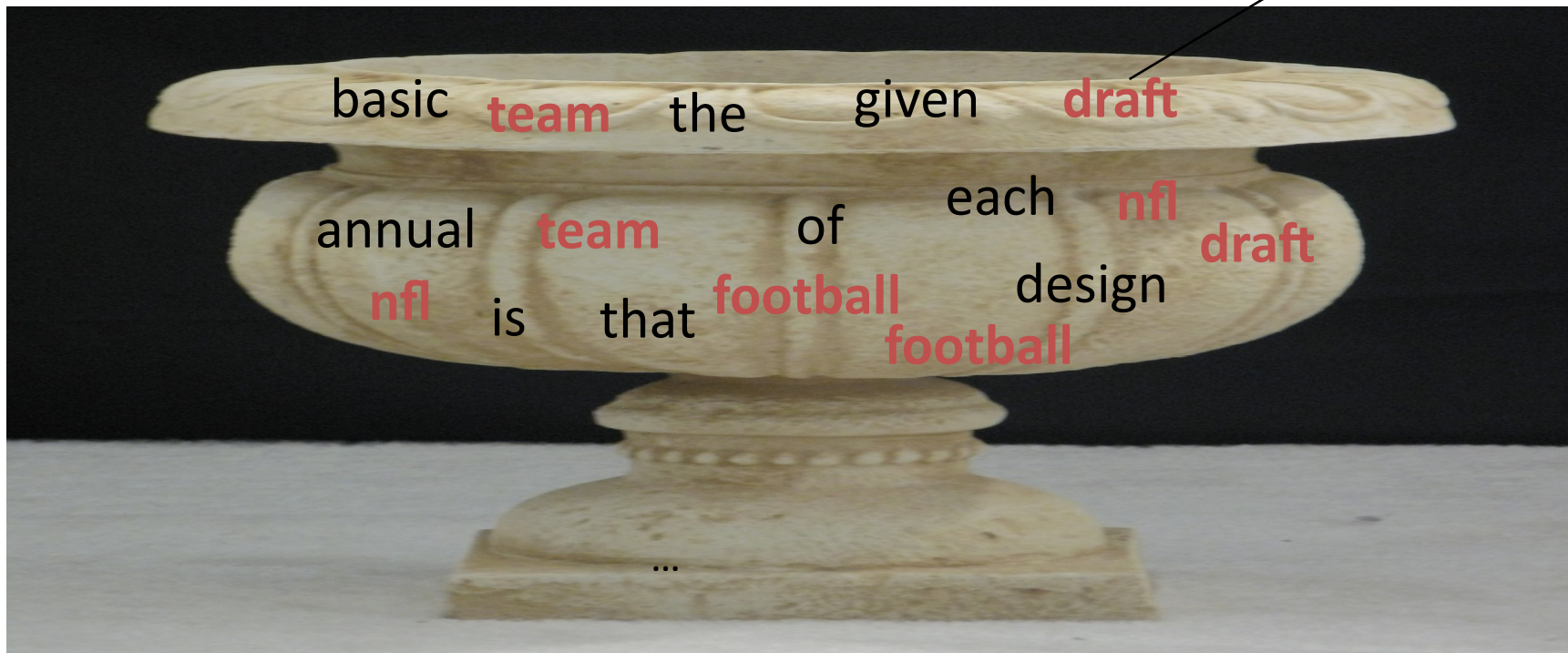
[Robertson et al. 1994, TREC City U.]

- BM25 “Best Match 25” (they had a bunch of tries!)
 - Developed in the context of the Okapi system
 - Started to be increasingly adopted by other teams during the TREC competitions
 - It works well
- Goal: be sensitive to term frequency and document length while not adding too many parameters
 - (Robertson and Zaragoza 2009; Spärck Jones et al. 2000)
- Belongs to the family of *probabilistic* IR models
 - The resulting formulae are close to tf.idf, but have a different foundation

Generative model for documents

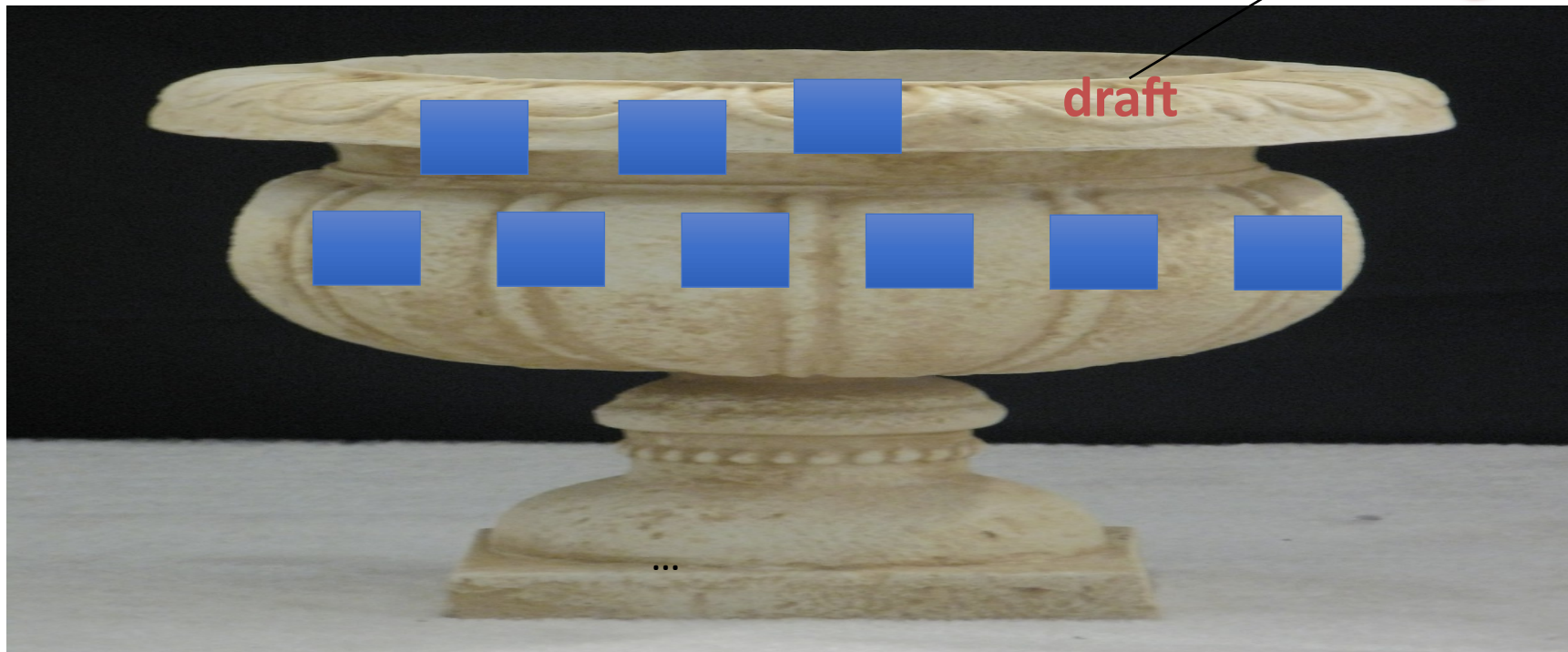
- Words are drawn independently from the vocabulary using a multinomial distribution

... the **draft** is that each **team** is given a position in the **draft** ...



Generative model for documents

- Distribution of term frequencies (tf) follows a binomial distribution – approximated by a Poisson



Poisson model

- Assume that term frequencies in a document (tf_i) follow a Poisson distribution
 - “Fixed interval” implies fixed document length ... think roughly constant-sized document abstracts
 - ... will fix later

Poisson distribution

- The Poisson distribution models the probability of k , the number of events occurring in a fixed interval of time/space, with known average rate λ ($= cf/T$), independent of the last event

$$p(k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

- Examples
 - Number of cars arriving at a toll booth per minute
 - Number of typos on a page

(One) Poisson Model flaw

- Is a reasonable fit for “general” words
- Is a poor fit for topic-specific words
 - get higher $p(k)$ than predicted too often

		Documents containing k occurrences of word ($\lambda = 53/650$)												
Freq	Word	0	1	2	3	4	5	6	7	8	9	10	11	12
53	expected	599	49	2										
52	<i>based</i>	600	48	2										
53	<i>conditions</i>	604	39	7										
55	<i>cathexis</i>	619	22	3	2	1	2	0	1					
51	<i>comic</i>	642	3	0	1	0	0	0	0	0	0	1	1	2

Eliteness (“aboutness”)

- Model term frequencies using *eliteness*
- What is *eliteness*?
 - Hidden variable for each document-term pair, denoted as E_i for term i
 - Represents *aboutness*: a term is elite in a document if, in some sense, the document is about the concept denoted by the term
 - Eliteness is binary
 - Term occurrences depend only on *eliteness*...
 - ... but *eliteness* depends on relevance

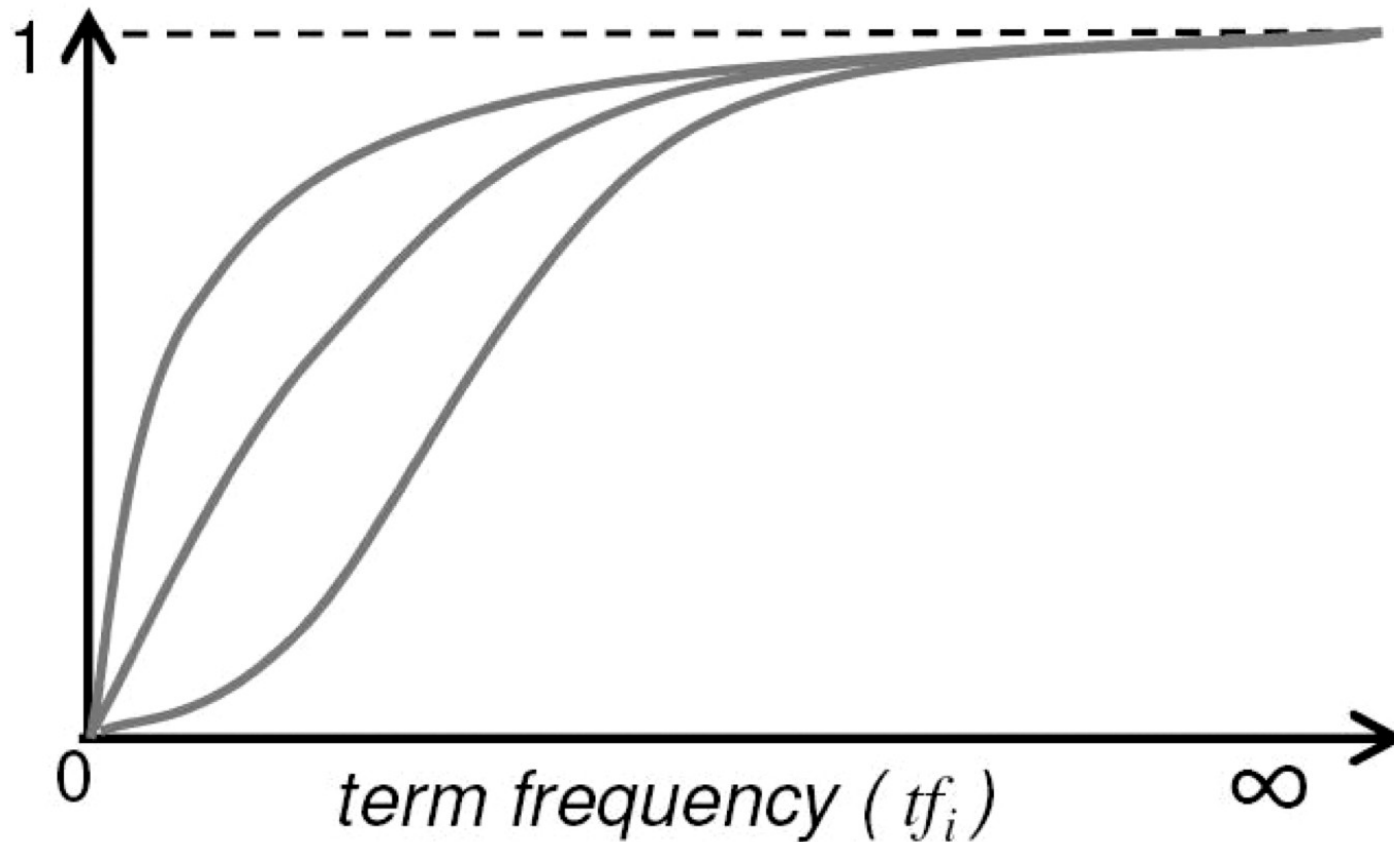
2-Poisson model

- The problems with the 1-Poisson model suggests fitting two Poisson distributions
- Document = mixture of ‘general’ terms and ‘topical’ terms
- In the “2-Poisson model”, the distribution is different depending on whether the term is elite or not

$$p(TF_i = k_i | R) = \pi \frac{\lambda^k}{k!} e^{-\lambda} + (1 - \pi) \frac{\mu^k}{k!} e^{-\mu}$$

- where π is probability that document is elite for term
- but, unfortunately, we don't know π, λ, μ

Let's get an idea: Graphing $c_i^{elite}(tf_i)$ for different parameter values of the 2-Poisson

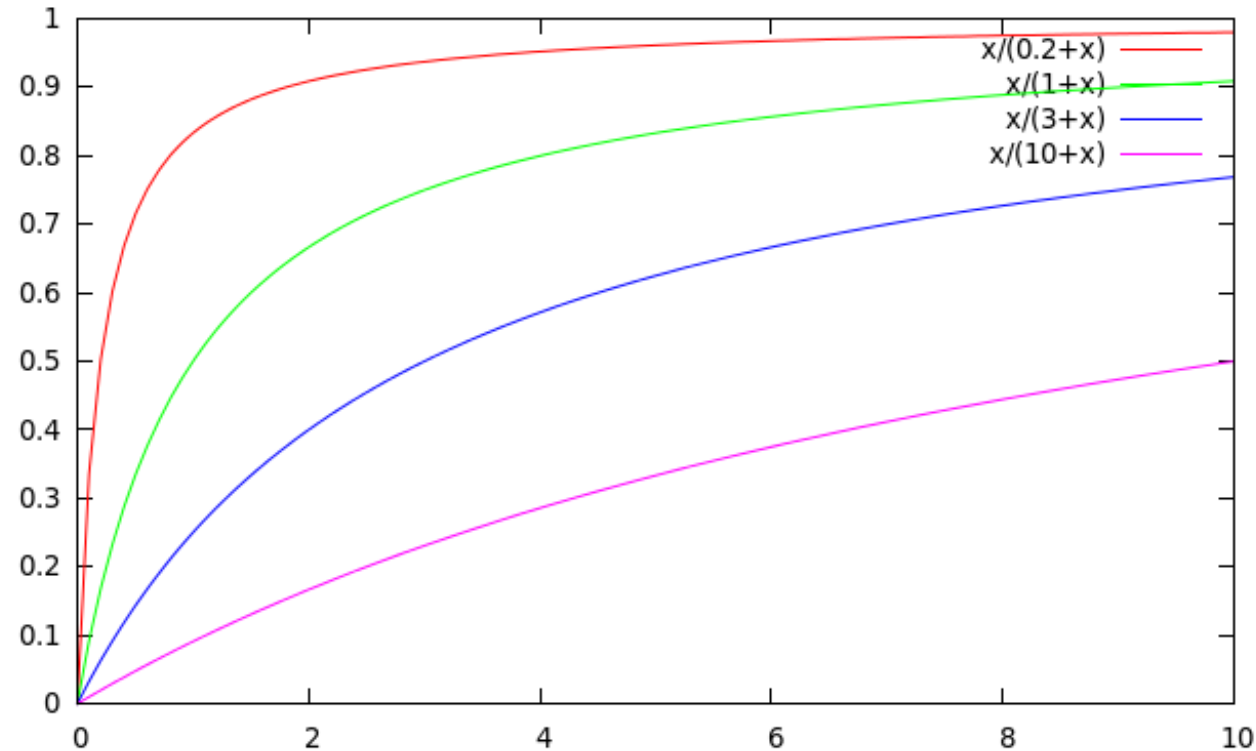


Approximating the saturation function

- Estimating parameters for the 2-Poisson model is not easy
- ... So approximate it with a simple parametric curve that has the same qualitative properties

$$\frac{tf}{k_1 + tf}$$

Saturation function



- For high values of k_1 , increments in tf_i continue to contribute significantly to the score
- Contributions tail off quickly for low values of k_1

Document length normalization

- Longer documents are likely to have larger tf_i values
- Why might documents be longer?
 - Verbosity: suggests observed tf_i too high
 - Larger scope: suggests observed tf_i may be right
- A real document collection probably has both effects
- ... so should apply some kind of partial normalization

Document length normalization

- Document length:

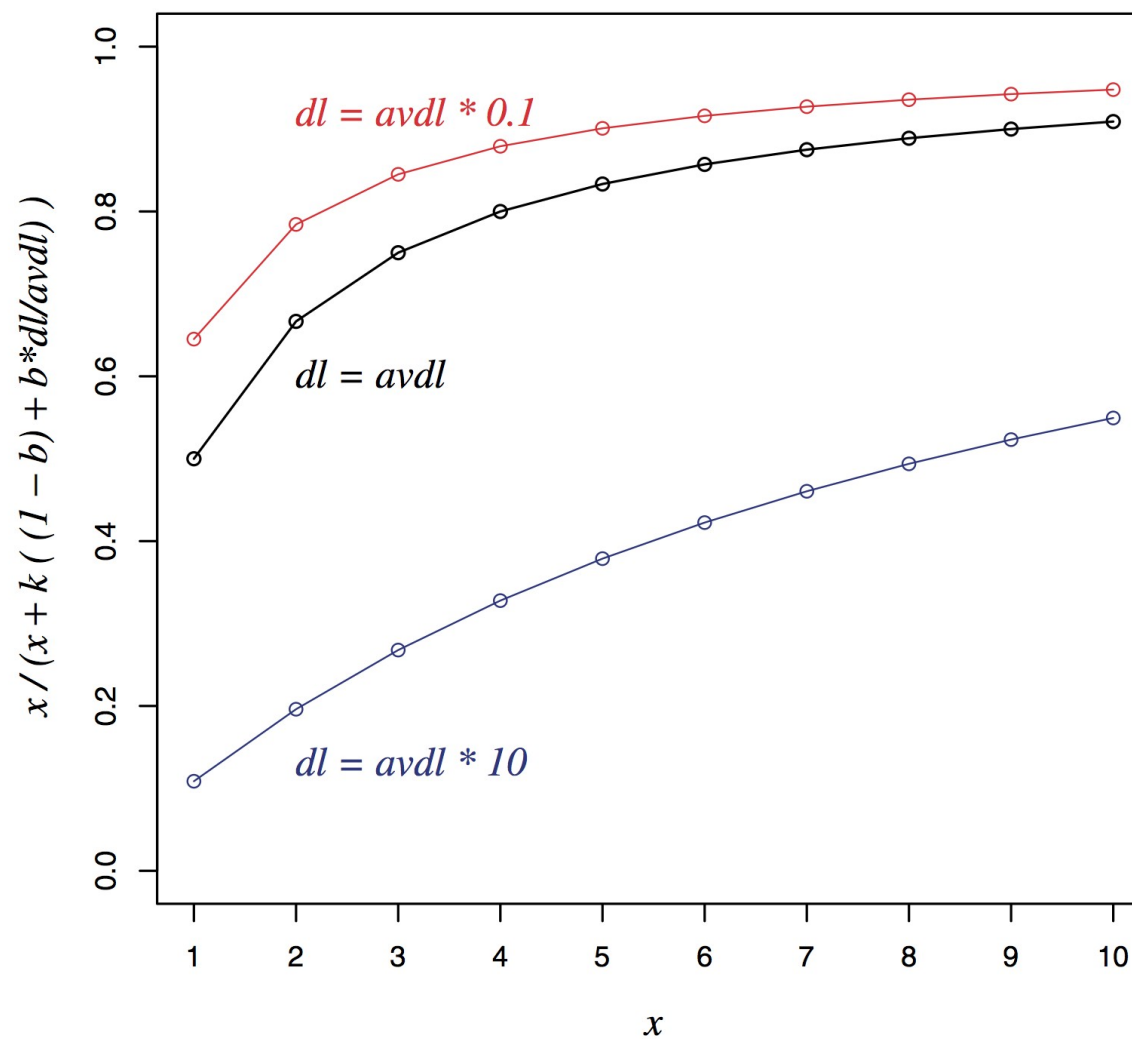
$$dl = \sum_{i \in V} tf_i$$

- *avdl*: Average document length over collection
- Length normalization component

$$B = \left((1 - b) + b \frac{dl}{avdl} \right), \quad 0 \leq b \leq 1$$

- $b = 1$ full document length normalization
- $b = 0$ no document length normalization

Document length normalization



Okapi BM25

$$RSV^{BM25} = \sum_{i \in q} \log \frac{N}{df_i} \cdot \frac{(k_1 + 1)tf_i}{k_1((1-b) + b \frac{dl}{avdl}) + tf_i}$$

- k_1 controls term frequency scaling
 - $k_1 = 0$ is binary model; k_1 large is raw term frequency
- b controls document length normalization
 - $b = 0$ is no length normalization; $b = 1$ is relative frequency (fully scale by document length)
- Typically, k_1 is set around 1.2–2 and b around 0.75

Why is BM25 better than VSM tf-idf?

- Suppose your query is [machine learning]
- Suppose you have 2 documents with term counts:
 - doc1: learning 1024; machine 1
 - doc2: learning 16; machine 8
- tf-idf: $\log_2 \text{tf} * \log_2 (N/\text{df})$
 - doc1: $11 * 7 + 1 * 10 = \mathbf{87}$
 - doc2: $5 * 7 + 4 * 10 = \mathbf{75}$
- BM25: $k_1 = 2$
 - doc1: $7 * 3 + 10 * 1 = \mathbf{31}$
 - doc2: $7 * 2.67 + 10 * 2.4 = \mathbf{42.7}$

References

S. E. Robertson and H. Zaragoza. 2009. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval* 3(4): 333-389.